

Questions			Answers	Marks
10	(a)	(i)	Progressive - there is energy transfer Longitudinal - the particle's displacement is parallel to the direction of transfer of energy of wave	B1 B1
		(ii)	$I = P/4\pi r^2$ $I = 20/4\pi(5^2) = 0.0637 \text{ W/m}^2$	C1 A1
		(iii)	Directional speakers are used in concerts because: • They focus energy toward the audience, making sound louder and clearer without increasing total power. • Energy waste is reduced – less sound spreads in unneeded directions (e.g., behind or upward). • Noise pollution is minimized – residential areas outside the beam receive less sound. • Better control over reverberation and feedback, especially in open-air environments.	B1
	(b)	(i)	Since <u>well-formed interference pattern</u> is obtained, the two sources are coherent.	B1
		(ii)	At the centre of the interference pattern, the <u>paths of the waves from both loudspeakers are equal</u> . Now since an <u>intensity minimum is formed (or total destructive interference)</u> there instead, it must mean that the speakers are generating waves in <u>anti-phase</u> .	B1 B1
		(iii)	$x = \frac{\lambda D}{a}$ $0.16 = \frac{\lambda(20)}{0.15}$ $\lambda = 1.2 \times 10^{-3} \text{ m}$ $f = \frac{v}{\lambda} = \frac{343}{1.2 \times 10^{-3}} = 2.86 \times 10^5 \text{ Hz}$	M1 C1

